



AIZG565

Machine Learning

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An overview of Gaussian Mixture Models

Mixture models



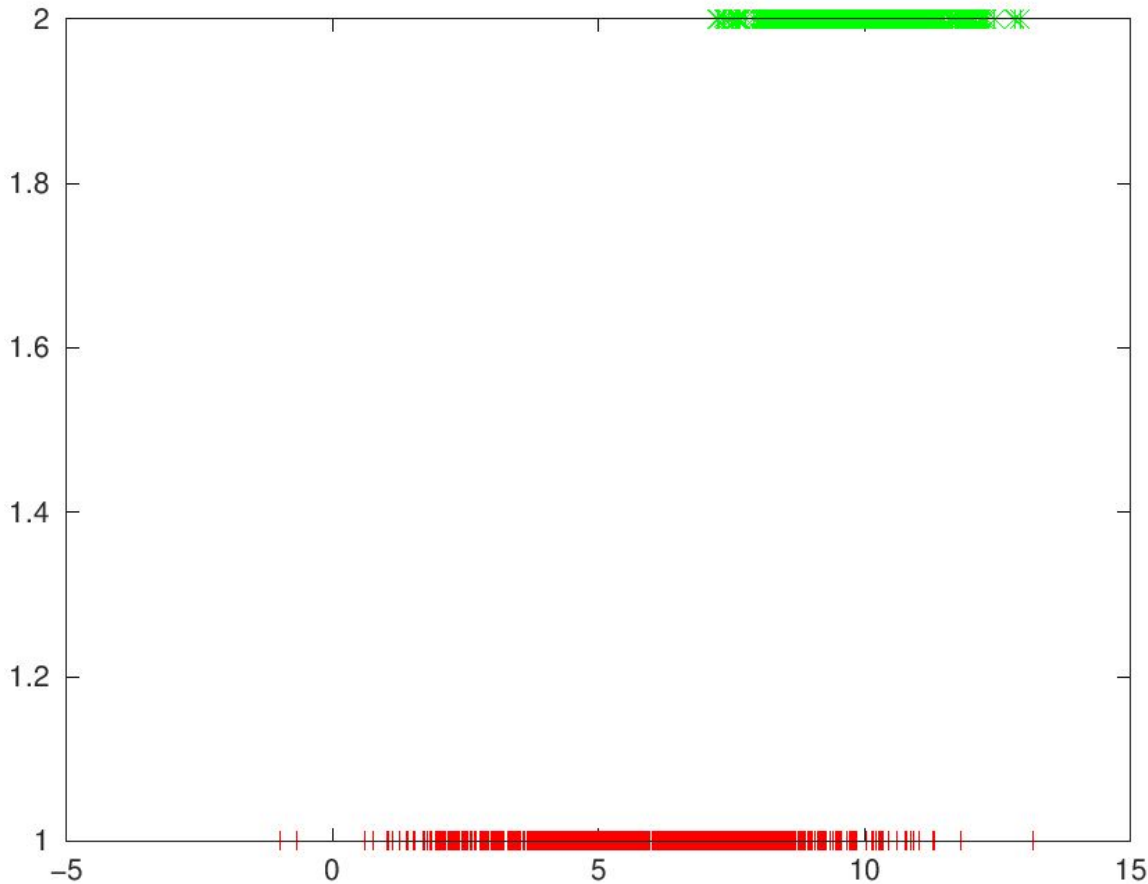
A mixture model typically models data as a weighted combination of several component distributions.

A common choice for component distributions is the Gaussian distribution. Such a mixture model is called a Gaussian Mixture Model (GMM) and the distribution is specified by the formula

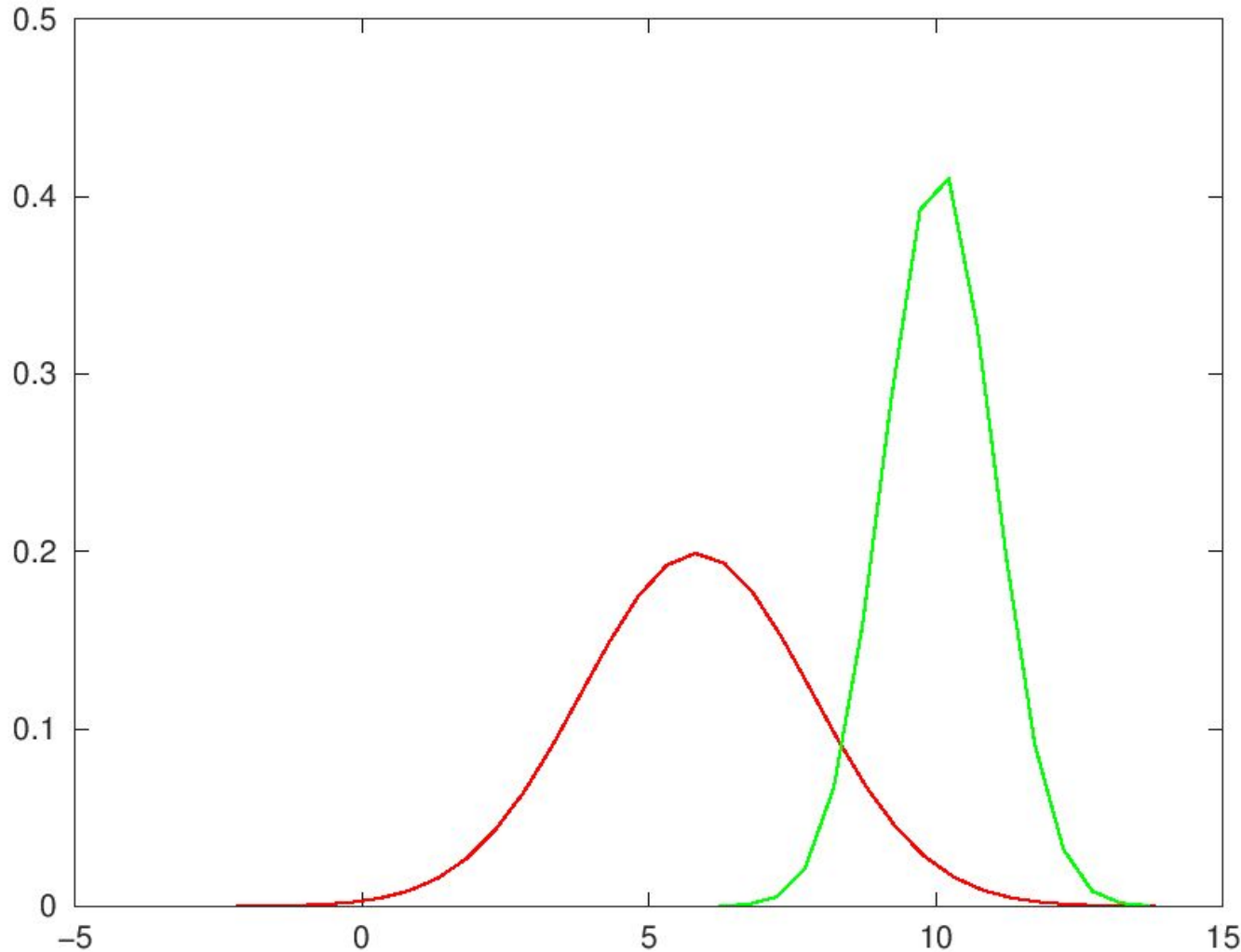
$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)$$

where, π_k s are the mixing proportions

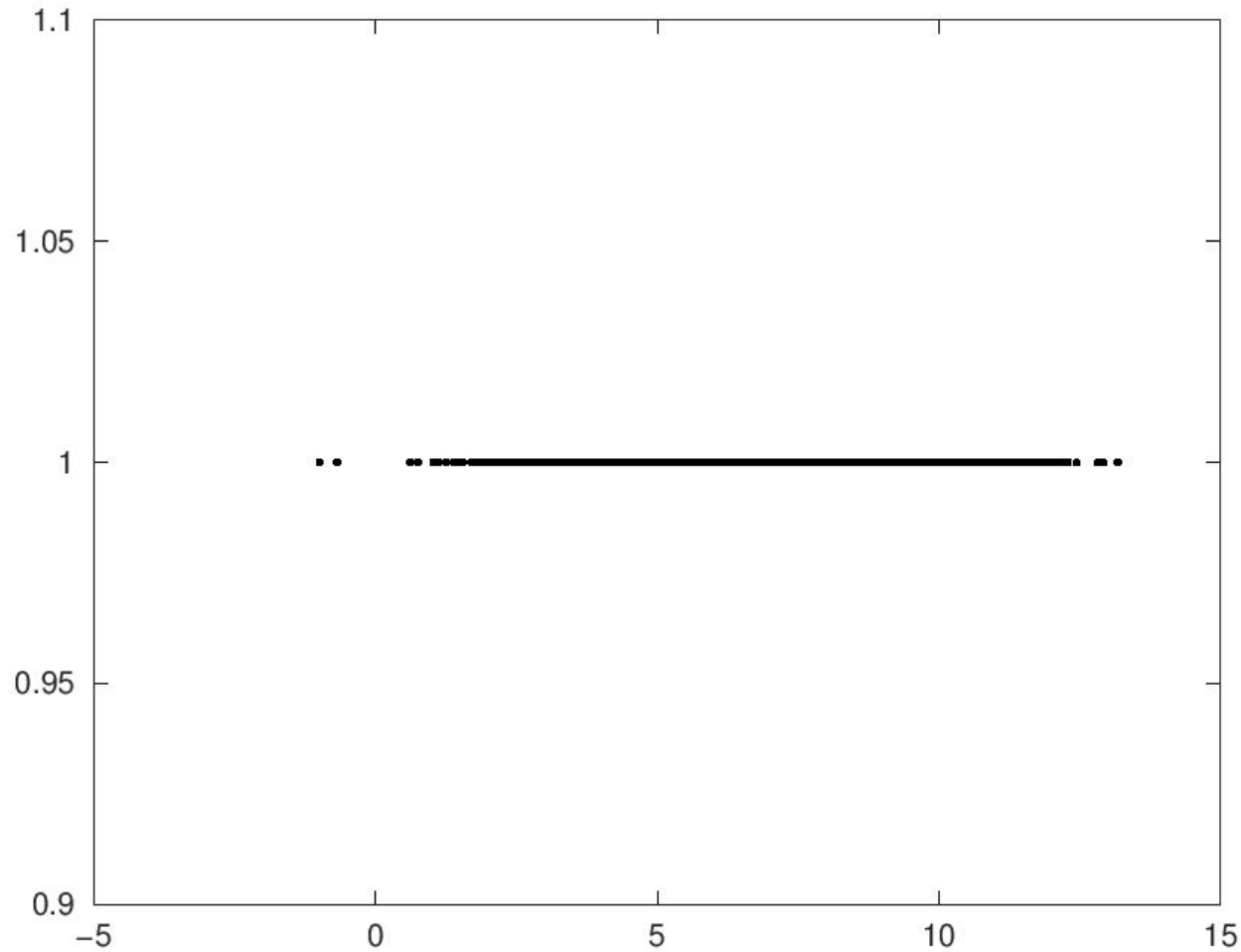
1D samples from 2 Gaussians



Gaussian densities estimated from the data



1D samples



Fitting a mixture of Gaussians



The most common approach for fitting a mixture of Gaussians is the **Expectation Maximization (EM) algorithm**. It alternates between two steps:

E-step:

Compute the posterior probability over z given our current model - i.e. how much do we think each Gaussian generates each datapoint.

M-step:

Assuming that the data really was generated this way, change the parameters of each Gaussian to maximize the probability that it would generate the data it is currently responsible for.

EM for GMM vs k-means



- EM for mixtures of Gaussians can be considered as a soft version of K-means, with fixed priors and covariance.
- In K-means, the assignments to clusters are 0 or 1 (hard assignments).
- In contrast, in the E-step, soft assignments are done, based on the softmax of the squared Mahalanobis distance from each point to each cluster.
- In EM, each center moved by weighted means of the data, with weights given by soft assignments

References



[Dempster, A.P.](#); [Laird, N.M.](#); [Rubin, D.B.](#) (1977). "Maximum Likelihood from Incomplete Data via the EM Algorithm". [*Journal of the Royal Statistical Society. Series B*](#). **39** (1): 1–38.

Expectation-Maximization (EM) Algorithm in ML Explained, March 9, 2026,
<https://www.appliedaicourse.com/blog/expectation-maximization-em-algorithm-in-ml-explained/>